

ABSTRACT

“House Price Prediction”

House price prediction is an important application of Machine Learning in the real estate industry. It helps buyers, sellers, investors, and real estate companies make informed decisions by estimating property prices based on factors such as location, number of bedrooms, bathrooms, living area, and house condition. Since manual price estimation can be difficult and time-consuming, this project uses Machine Learning techniques, specifically Linear Regression and Gradient Boosting Regressor, to predict house prices accurately. Linear Regression establishes relationships between house features and prices, while Gradient Boosting Regressor improves prediction accuracy by combining multiple decision trees. These algorithms learn patterns from historical housing data and provide reliable price predictions, making them suitable for real estate applications.

Python is used as the main programming language in this project because of its flexibility and powerful machine learning libraries. Pandas is used for data loading, cleaning, and manipulation, while NumPy supports numerical computations. Matplotlib and Seaborn are used for data visualization and analysis. The housing dataset is preprocessed by removing unnecessary columns, handling missing values, selecting important features, and preparing the data for model training. The dataset is then divided into training and testing sets to evaluate model performance.

This project demonstrates how Machine Learning can be used to predict house prices automatically and accurately. It provides an efficient approach for property valuation and real estate analysis. Using Python and its machine learning libraries, the project shows how predictive models can solve real-world problems in the housing sector. It also provides a foundation for future improvements, such as integrating real-time market data, using deep learning techniques, and developing web-based prediction systems.

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